

Multiple Hypothesis Tracking Using Neutral-Atom Quantum Computers

A hybrid classical–quantum pipeline for data association

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1 Problem

A sensor returns **unlabelled** detections at every scan. Nothing says which target produced which detection, some targets are missed, and some detections are clutter. Committing early is dangerous: if two targets cross and their identities swap, the error propagates forever.

Multiple Hypothesis Tracking (MHT) keeps many competing explanations alive and defers the decision. The price is combinatorial growth, and that is where a quantum neutral atom array could help.

2 From MHT to MWIS

Each candidate track becomes a vertex weighted by its likelihood score, and an edge joins two tracks that claim the same detection. a type of problem called **Maximum Weight Independent Set (MWIS)**. The best global hypothesis is then

$$\max \sum_i w_i x_i \quad \text{s.t.} \quad x_i + x_j \leq 1 \quad \forall (i, j) \in E.$$

3 Rydberg physics

The MWIS problem maps naturally onto neutral atoms, whose Rydberg blockade mechanism enforces independence of solutions.

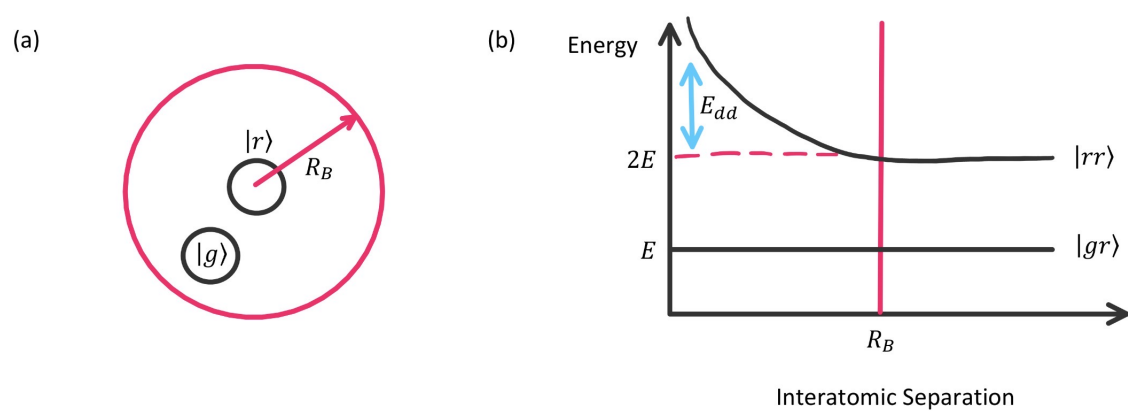


Fig. 1 (a) Rydberg blockade under a global drive. The atom outside the blockade radius R_B is excited; the one inside is not. (b) Within R_B the dipole–dipole interaction E_{dd} light-shifts the unexcited atom off resonance

As each candidate track is encoded as an atom, placing incompatible tracks within R_B therefore enforces $x_i + x_j \leq 1$ in hardware. Driving on resonance ($\delta=0$), the Rydberg system maps exactly onto a QUBO Hamiltonian:

$$H = \frac{1}{2} \sum_i \Omega_i \sigma_i^x - \sum_i \delta_i n_i + \sum_{i < j} \frac{C_6}{r_{ij}^6} n_i n_j$$

Weights are encoded as local detunings δ_i , applied per atom, to vary their blockade radii. A slow sweep of the Rydberg drive exploits the Quantum Adiabatic Theorem to evolve the prepared ground state into the ground state representing the solution to the problem.

Encoding a conflict graph on the array

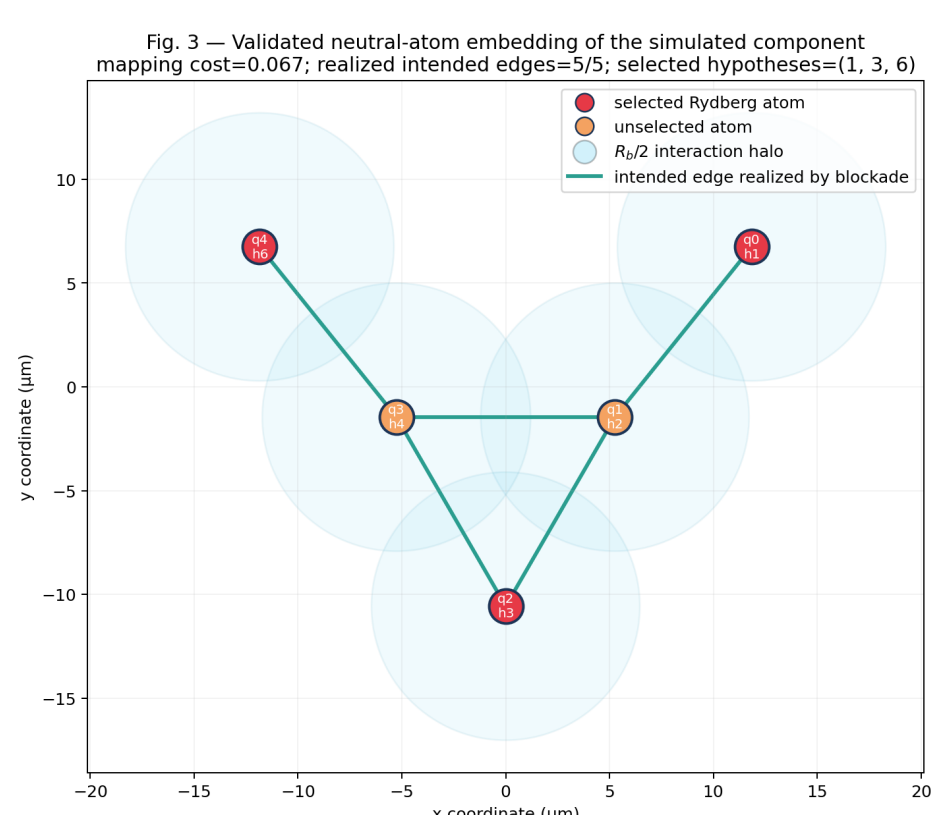


Fig. 2 A validated embedding of one conflict component. All five intended edges are realised by the blockade, and the atoms left in the Rydberg state at the end of the sweep name the selected hypotheses, here (1, 3, 6), matching the exact classical optimum.

4 Hybrid workflow

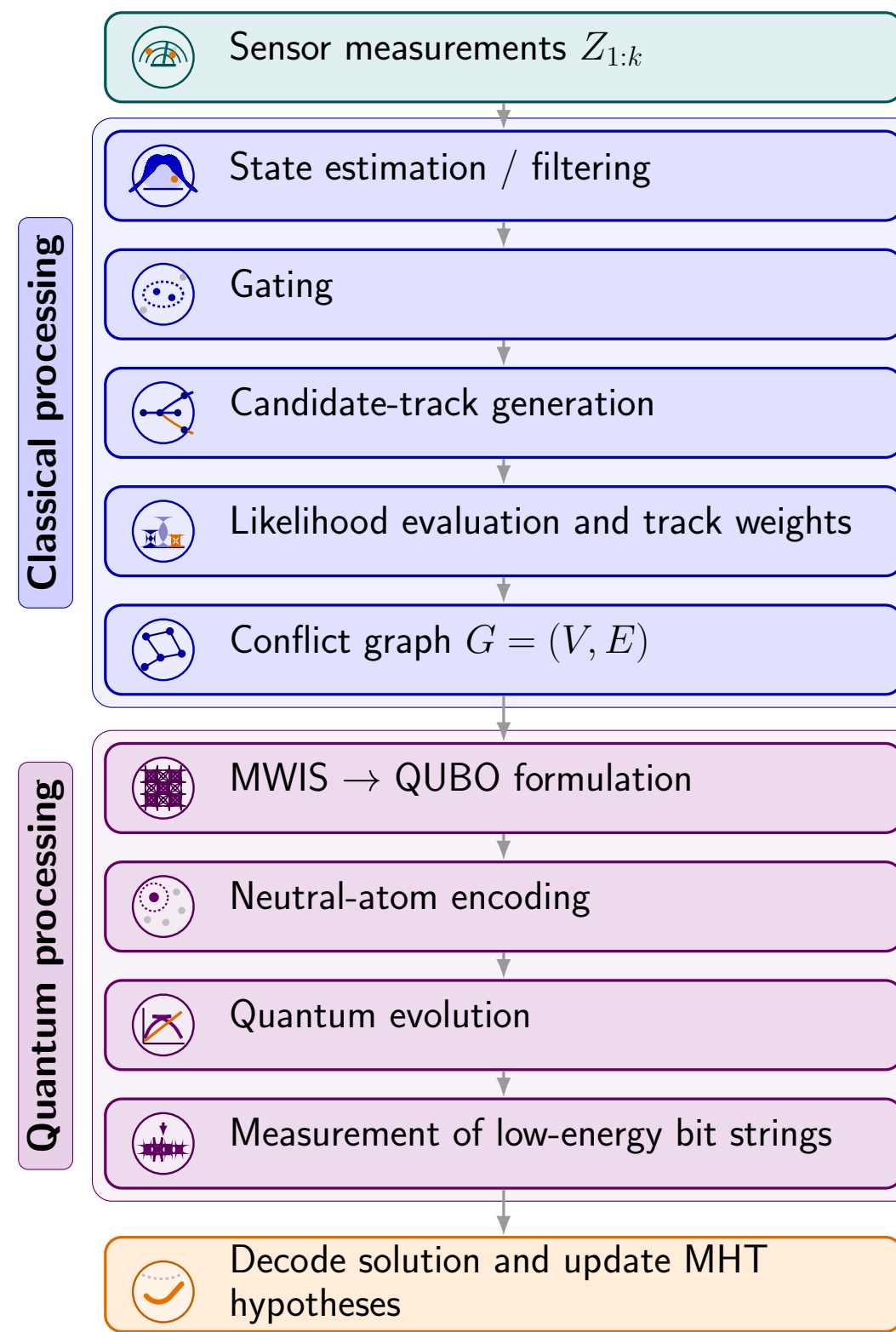


Fig. 3 Classical processing constructs and scores candidate track hypotheses, which are converted into a conflict graph and then mapped to an MWIS/QUBO problem. The quantum stage encodes and evolves the optimisation problem, after which low-energy bit strings are measured and decoded to update the MHT hypotheses.

5 Data and detection

Two data sources exercise the same pipeline. **RADIATE** supplies real Navtech CTS350-X radar, 360° range–azimuth at 4 Hz, from the *tiny_foggy* sequence; synthetic generators supply matched controlled scenarios with known ground truth, as points and as images.

Detection was validated against noise. Precision falls steadily as noise rises, consistent with the detector reading noise as signal, while recall shows no clear trend: the detection step consistently misses certain objects. Quantum performance was **classical adiabatic optimization**, to assess quantum advantage.

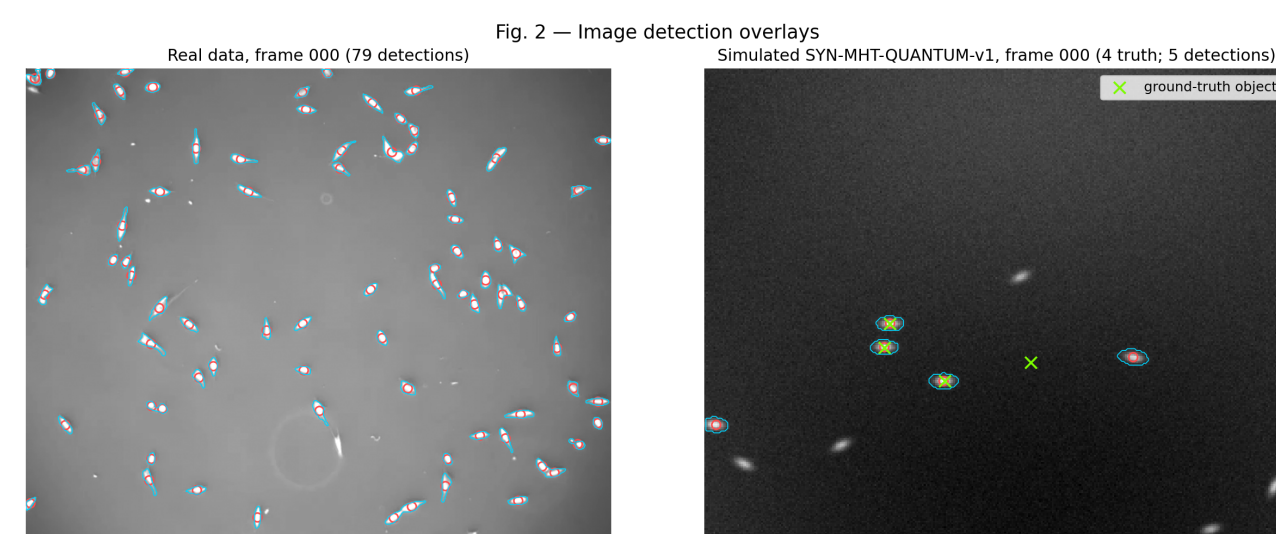


Fig. 4 Image detection overlaid on cell image data (left) and synthetic data (right).

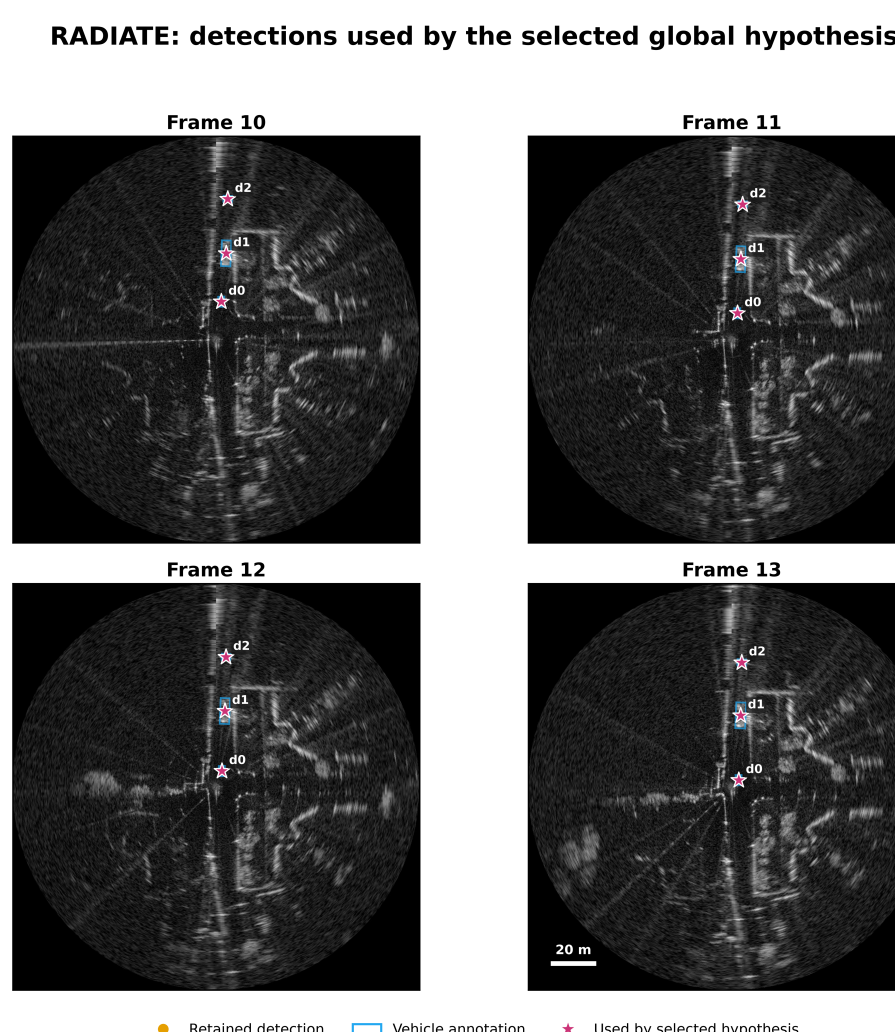


Fig. 5 The pipeline on real radar: four consecutive *tiny_foggy* frames with the detections used by the selected global hypothesis, over the raw Navtech returns.

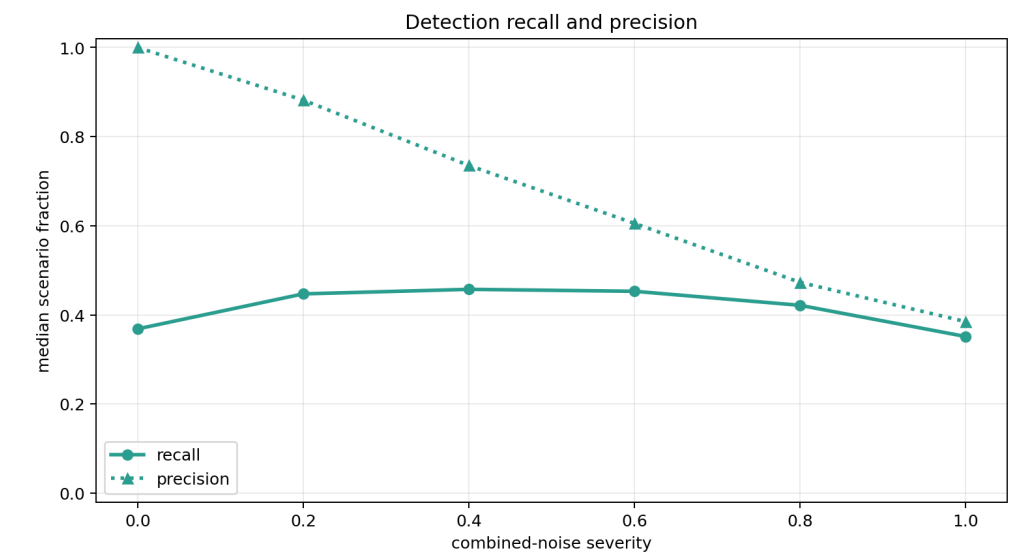


Fig. 6 Detection and tracking quality against combined-noise severity, median over 20 object-count and 100 randomized trials. Measured by precision and recall showing the rate of false positives increasing with noise.

6 Quantum reliability

Sparsity is not enough. Consider a **star graph**: one central vertex, many leaves. It is sparse, yet the spacing between neighbouring leaves at fixed radius shrinks as the leaf count grows, so atoms representing vertices that share *no* edge can be forced close enough to interfere. The Rydberg interaction is tunable but not free, so the problem persists and some graphs cannot be fully embedded.

Across both panels, **whenever the graph is successfully implemented the optimal solution is found**: no completed-but-suboptimal case was observed anywhere. Increased noise severity lowers the number of successful embeddings at fixed size $n = 18$, as noisier data carries more low-likelihood competing hypotheses for the geometry to trip over. Increasing vertex count at fixed low noise does the same.

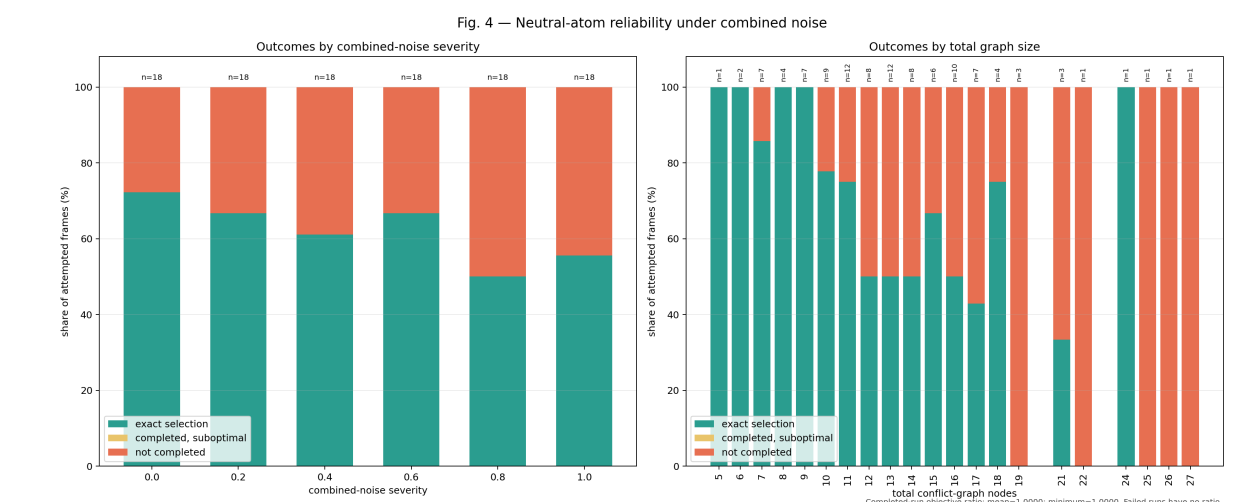


Fig. 7 Outcomes of the neutral-atom adiabatic approach on synthetic data of varying noise level (left) and graph size (right).

7 Runtime

Quantum runtime was estimated from simulation data converted to hardware time using Pasqal pulse and measurement parameters. Even on limited classical hardware the **classical device vastly outperforms the quantum process**, and normalising classical power to match the quantum runtime at five vertices shows **no discernible scaling advantage**.

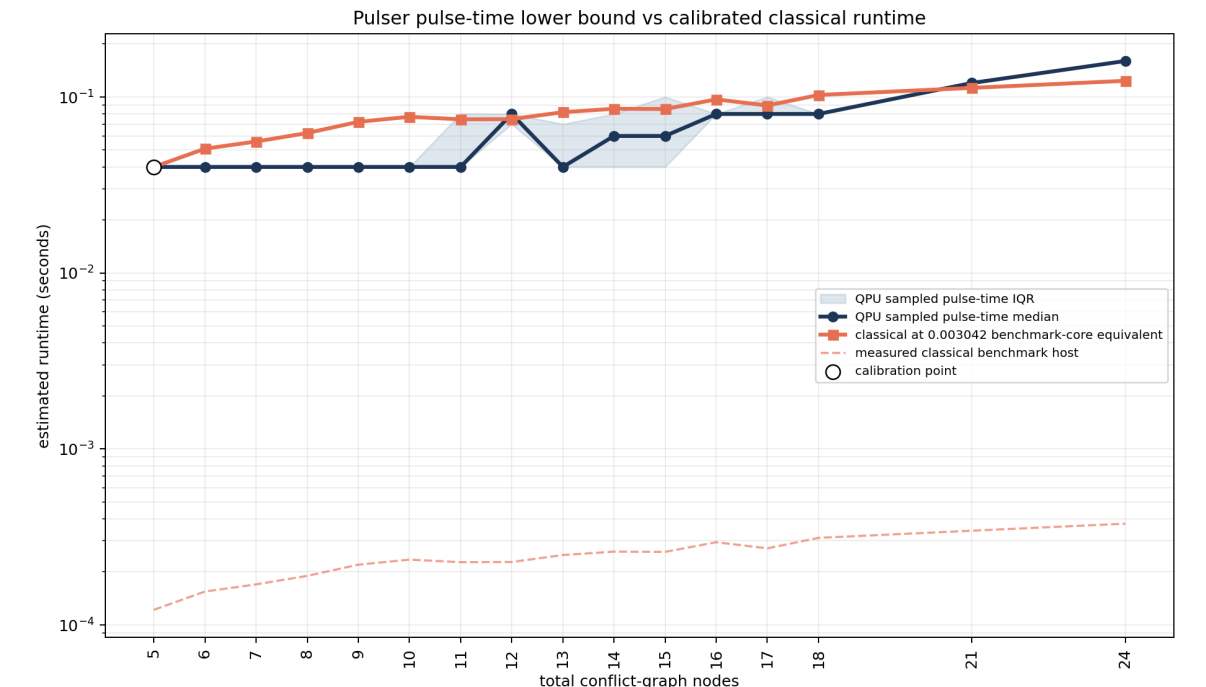


Fig. 8 Classical MWIS solver runtime against the estimated quantum runtime. A CPU scaled to match the quantum runtime at five vertices is shown to expose scaling behaviour.

8 Conclusion

At first glance MHT via the MWIS formulation suits a neutral-atom quantum adiabatic algorithm well, but there are significant complications.

- **Geometry dominates.** Embedding the graph without introducing unintended edges needs further work; residual cross-talk also limits accuracy.
- **The feedback loop is costly.** Every frame needs classical pre-processing, embedding optimisation and repeated shots, all of which must interface back to the classical device before the next frame.
- **Classical methods already suffice here.** Gating, pruning and clustering let simulated annealing succeed at every scale we could simulate. It may degrade at larger ones.

References. [1] Bernot et al., arXiv:2506.13228 (2025). [2] Farhi et al., quant-ph/0001106 (2000). [3] Papageorgiou & Salpukas, Springer (2009) 235. [4] Sheeny et al., *RADIATE*. [5] Silvério et al., *Quantum* **6**, 629 (2022).